

Peer Review

Review of: "Cyanobacterial Biomass Estimation in Subtropical and Temperate Reservoirs by Sentinel-2"

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Suggestions and comments

General comment

The work is a good contribution to using remote sensing aimed at the identification of potentially toxic cyanobacteria in two distant and distinct basins. Their methodology allows for reproducible future studies. The authors present their best findings, pondering some pitfalls and caveats. They discussed fairly with the specific literature and offered concluding remarks.

Specific comments

For the **Abstract**: "The study of cyanobacterial biovolume, a relative abundance of phytoplankton in terms of biomass, using remote sensing is not common. Its estimation can be done in two ways: indirectly, through the determination of phycocyanin concentration, since it is their main pigment; or directly, by determining their abundance or biovolume. This work aims to improve expertise in the direct estimation of cyanobacterial biovolume using Sentinel-2 imagery. To empirically develop the algorithm, 43 georeferenced samples were collected: 20 from the subtropical zone and 23 from the temperate zone, for which cyanobacterial biovolumes were determined. Images were resampled and atmospherically corrected with the Case2 eXtreme neural net, and the remote sensing reflectance values were extracted. The best results were obtained with the NDB5B4, yielding an R^2 of 0.92, an RMSE of $1.71 \text{ mm}^3 \text{ L}^{-1}$, and an NRMSE of 7.6%. To demonstrate its consistency and functionality, the algorithm was applied to images collected over two years to reproduce both temporal and spatial variations in cyanobacterial biomass. The highest values were recorded in the subtropical zone. Tail-dam asymmetry was observed in both

climatic zones. In the temperate zone, the annual variation showed both a minimum and a maximum annual value, whereas more variability was observed in the subtropical zone.”

I also suggest the authors insert some *concluding remarks* as a final statement in the Abstract.

Introduction

The authors gave some long statements. I recommend the authors split those long statements, for example, by developing better examples of *driving force* (anthropic influence, pollution) and the association with the increase of *harmful cyanobacteria blooms*. Also, instead of using HAB (harmful algal blooms), which is more general and comprehensive, the authors can adopt specifically *harmful cyanobacteria bloom* or *harmful cyanobacteria blooms* throughout the text.

First paragraph, Page 2: I guess the authors could explore better each driving force, offering a kind of effect over the phytoplankton community/ecosystem. Instead of presenting all of them in a unique, long sentence, I recommend the authors dismantle that statement and build one to two statements for that rationale.

In general, I strongly recommend the authors rephrase their statements by adopting verbs in *past tense* and/or modal verbs, mainly from their Results up to Conclusions.

In relation to their Tables, I also recommend those tables have sides opened. Contrarily, those would be Boxes.

Figure caption in Figure 1 - I recommend writing each and every full name of the reservoir (meaning of the acronyms) and the lake in the figure caption.

Figure caption in Figure 2 - Indicate the meanings for the names on the map: municipalities? water bodies?

Page 12, first paragraph... An example of adopting *past tense* verbs.

"The clouds **modified**... and each reservoir **had** a different..."

Caption for Table 4 - Adopt sides opened for being a table. Insert into the figure caption "data = used data (transformed values with Log10 or square root)."

Image output

First paragraph - This part is more appropriate for the Materials and Methods section.

Show only your results/findings here.

Page 16 - I suggest replacing *heaved* with *did not show* coincidences.

Page 17, Last paragraph - Explanation here is not adequate...

This explanation should be appropriate for Materials and Methods and/or Discussion.

Table 5 - Rebuild the table with sides opened. And put in this caption "n.s. = non-significant ($p > 0.05$)."

Figure 8 - The value of the end part of the colour bar (right up corner) is missing.

Page 22, first paragraph - "...would be the overestimation of biovolume values upper that threshold..."

Attachments: available at <https://doi.org/10.32388/VWW9CY>

Declarations

Potential competing interests: No potential competing interests to declare.